THE COMPETITION: INDIVIDUALS AND SOCIAL STRUCTURES WITH MUSICAL AI

Daniel Pitman
Elder Conservatorium
The University of Adelaide

ABSTRACT

This project is a listening room installation that treats multiple instances of neural networks as individual agents, and explores emulating 'social structures' to refine solutions to musical problems. Populations of 'individual' agents are organised into various groups such as a panel of judges, bands, and an audience. These groups will refine a 'best' possible solution, live, for specific physiological challenges defined beforehand.

This installation was not the original aim of the research project that gave rise to the agents. Originally neural networks were created and trained using a music algorithm and a physiological feedback from a selection of human participants to drive the algorithm 'solo' for composition tasks (links to examples are included). Due to the large numbers of variables involved and the broad range of responses from the participants, the confidence of the networks is typically short of being mathematically satisfactory for any strong conclusions which is fairly common in affective algorithmic composition systems (Williams et al. 2014).

However, for musical/creative agent outcomes this is satisfactory, as it is difficult to argue that any musical structure is „incorrect‟ for any given purpose. It was found that the training process of these neural networks led to situation where each instances could contribute unique but technically „correct‟ musical solutions to an affective target with similar confidence, leading to the conclusion that these networks had unique approaches to interpreting the reactions of humans to changing musical structures.

An affective challenge will be set requiring a musical passage which can attempt to provoke a particular physiological reaction in humans as a solution. The agents are arranged into three groups. First is the panel of judges, best trained of all the agents, and prone to mild audio gestures. The second is the bands themselves, who perform generated pieces. Third is the audience, who provide approval or disapproval as applause, yelling, and other gestures. These don‟t affect the songs generated, but have a chance to influence the judge‟s decisions. While all the agents are trained using the same data and architecture, the nature of training neural networks imposes unique results for each agent, which informs their respective decisions.

The public will be welcome to enter the listening room and listen to the progress of the competition using a small surround sound PA or binaural headphones. The bands will play towards a winner for the duration of the time allotted, and the best scoring pieces will be combined into one piece, and can be presented at the end of the conference either online, and/or in a short presentation.

Algorithmic affective works for piano generated using similar agents can be heard via the following links:

https://soundcloud.com/danpitman/searching-for- machine-the-affective-algorithmic-composer

https://soundcloud.com/danpitman/good-vs-evil-the- affective-algorithmic-composer

https://soundcloud.com/danpitman/moments-in-the-mathematical-forest-the-affective-algorithmic-composer

REFERENCES

Williams, D., A. Kirke, E. R. Miranda, E. Roesch, I. Daly, and S. Nasuto. 2014. "Investigating Affect in Algorithmic Composition Systems." Psychology of Music, August.